

Estimating the Installed Capacity of Aggregated Thermostatically Controlled Loads from Net Load and Temperature

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Abstract—The electrification of heating and cooling is connecting more thermostatically controlled loads (TCLs), such as heat pumps (HPs), to low-voltage networks. However, distribution system operators (DSOs) rarely know the installed capacity of the TCLs behind each substation or feeder. This paper estimates that capacity from the aggregated net load and ambient temperature that DSOs already hold. The installed capacity follows as the ratio of two temperature sensitivities: that of a piecewise-linear net-load fit, and that of a simultaneity factor (SF) fit calibrated on a submetered pilot. Only the SF sensitivity is transferred; each aggregate keeps the threshold temperature of its own net-load fit. On 1260 semi-synthetic aggregates of Swiss households, the capacity is recovered at a weighted absolute percentage error (WAPE) of 19%, with the error growing as the share of households with an HP falls. On three real Swiss and German aggregates outside the pilot, where every household has an HP, the estimates lie within 10% of the submetered capacity, against up to 26% when the full SF curve is transferred. The temperature-driven HP energy needs no pilot and is recovered at a WAPE of 9%, and a flexibility potential derived from both fits at a WAPE of 10% against its submetered reference. Across populations from five countries, the SF sensitivity transfers within 12% inside one climate zone. It differs by 13–14% between HP technologies, however, and up to 1.7-fold between climate zones and between social and private housing.

Index Terms—Thermostatically controlled loads, heat pumps, low-voltage networks, simultaneity factor, installed capacity, smart meter data, flexibility, distribution system observability.

NOMENCLATURE

DER	Distributed energy resource.
DSO	Distribution system operator.
HP	Heat pump.
IQR	Interquartile range.
LV	Low voltage.
NILM	Non-intrusive load monitoring.
OLS	Ordinary least squares.
PV	Photovoltaics.
SF	Simultaneity factor.
TCL	Thermostatically controlled load.
WAPE	Weighted absolute percentage error.
P_{net}	Aggregated net load.
P_{TCL}	Aggregated TCL load.
$P_{\text{non-TCL}}$	Aggregated non-TCL load.
P_{base}	Temperature-independent base load.
$\hat{P}_{\text{TCL}}^i$	Fitted TCL load of aggregate i .
s_h, s_c	Heating, cooling sensitivities of the net-load fit.
T_h, T_c	Heating, cooling threshold temperatures.
$P_{\text{TCL}}^{\text{max}}$	TCL installed capacity.
P_k^{max}	TCL installed capacity active in regime k .
SF_h, SF_c	Heating, cooling simultaneity factor.
$\hat{\text{SF}}_h, \hat{\text{SF}}_c$	Fitted heating, cooling simultaneity factor.
b_h, b_c	SF intercepts at the threshold temperatures.
m_h, m_c	Heating, cooling SF sensitivities.
R_h^2, R_c^2	Coefficient of determination of a fit on the days of the heating, cooling regime.
α_F	Flexibility duty cycle.
Δt	Energy integration period.

Δt_{Th}	Thermal-inertia window.
E_{TCL}	Temperature-driven TCL energy.
E_{flex}	Flexible energy.
N, N_{hp}	Number of consumers and heat pumps.

I. INTRODUCTION

The electrification of residential heating and cooling is a central lever of decarbonisation policy. Its instruments are thermostatically controlled loads (TCLs), such as heat pumps (HPs), electric resistance heaters and air conditioners, and the HP leads their uptake [1]. With them grows the share of household demand that follows outdoor temperature. On 200 German low-voltage (LV) feeders, for example, installed HP power rose by over twenty per cent from 2023 to 2025 [2].

This uptake supports decarbonisation, yet it loads LV networks that were not sized for it. Field and simulation studies show that TCLs raise coincident feeder and transformer peaks in extreme weather. For HPs, the most studied case, these peaks are sharpest in cold weather and add to those of electric vehicles and rooftop solar [3]–[7]. To plan and operate these networks, network operators must know how much thermostatic load sits behind each metering point.

A distribution system operator (DSO) rarely holds that information. It meters aggregated net load and can obtain weather data. It usually knows how many consumers each feeder serves, but seldom holds the appliance registers or nameplate ratings needed to interpret the thermal response [2], [8]. The installed capacity of the TCLs behind a feeder therefore stays largely unobserved.

Installed capacity, rather than the coincident peak, is the quantity sought here, for three reasons. It scales with adoption, so it supports planning scenarios as the number of TCLs grows. It bounds the peak the population can draw when its devices are synchronised, for instance after a tariff blocking period, in which the utility interrupts the HP supply, or during the activation of flexibility. Finally, multiplied by the simultaneity factor (SF) used in planning, i.e. the share of the installed capacity drawn at once, it gives the design peak that sizes transformers and cables.

Five lines of research approach this gap, summarised in Table I. Non-intrusive load monitoring (NILM) and HP detection recover thermostatic load from customer meters, either as a present-absent label [9], [10], [34] or as a disaggregated energy time series [11], [12]. Some extend to LV substations [14], [15]. These methods, however, need labelled or submetered training data, stop short of the installed capacity, and remain largely synthetic at the substation.

Change-point and degree-day models fit load as a piecewise-linear function of temperature. They range from canonical building-level formulations [35]–[37], reviewed in [38], to aggregated and smart-meter populations [17]–[19]. They recover heating and cooling sensitivities and a threshold temperature without any labels. They stop, however, at the coincident response and never reach the non-coincident installed sum.

For photovoltaics (PV) and other distributed energy resources (DERs), by contrast, installed-capacity estimation is well developed.

TABLE I
PRIOR WORK ON DETECTING AND QUANTIFYING DERS AND TCLS

Reference	DER	Task	Level	Method	Input	Installed capacity	Submetering needed		
							Target	Training	Transferable
<i>Detection and disaggregation</i>									
[9]	HP	Detection	Meter	ML (PU learning)	Smart meter			✓	
[10]	HP	Detection	Meter	ML	Smart meter + weather			✓	
[11]	HP	Disaggregation	Meter	ML (deep)	15-min smart meter			✓	
[12]	HP	Detection + disaggregation	Meter	ML (CNN)	Smart meter			✓	
[13]	PV/EV/HP	Detection	Meter	ML	Net demand			✓	
[14]	Loads/DER	Regional NILM	Substation	NILM	Substation load			✓	
[15]	Loads	Disaggregation	Substation	ML (deep)	Substation load			✓	
[16]	AC	Real-time disaggregation	Feeder	Online learning	Feeder load + device models			✓	
<i>Temperature-response models</i>									
[17]	Cooling	Change-point model	Community	Change-point	Aggr. load + temp.				
[18]	HVAC	Change-point model	Meter	Change-point	Smart meter + temp.				
[19]	Sector load	Weather effects	System	Regression	Sector load + weather				
<i>TCL aggregate modelling</i>									
[20]	TCL	State estimation, control	Population	Markov model + Kalman filter	Aggr. load + TCL model			✓	
[21]	TCL	Flexibility (virtual battery)	Population	Physics (thermal)	TCL parameters + temp.			✓	
<i>PV and transformer capacity</i>									
[22]	PV	Detection + capacity	Meter	ML	Smart meter	✓		✓	
[23]	PV	Capacity + energy	Meter	ML	Smart meter	✓		✓	
[24]	PV	Disaggregation	Feeder	Physics	Power flow + GHI				
[25]	PV	Capacity	Feeder	Probabilistic	Aggr. net load	✓			
[26]	PV	Capacity	Substation	ML / model-based	Aggr. net load + GHI	✓		✓	✓
[27]	Transformer	Nameplate rating	Transformer	Model-free	Smart meter	✓			
<i>HP demand, coincidence and flexibility</i>									
[6]	PV/EV/HP	Coincidence / impact	Feeder	Empirical	Smart-meter profiles				
[28]	HP	Coincidence / ADMD	Population	Field study	Submetered HP		✓		
[29]	HP	Peak in cold weather	Population	Field study	Submetered HP		✓		
[30]	HP	Demand vs temperature	Population	Probabilistic	Submetered HP + temp.			✓	✓
[31]	HP	Load profiles	Feeder	Physics (thermal)	Temp. + housing				✓
[32]	HP	Flexibility / DR	Population	Field study	Submetered HP		✓		
[33]	Appliances	Flexibility / DR	Population	Field study	Submetered appliances		✓		
This work	HP / TCL	Capacity, energy, flexibility	Aggregate	Change-point + SF	Net load + temp. + pilot SF	✓		✓	✓

Installed capacity: recovers the installed capacity. Submetering needed: submetered device data or labels are needed on the target aggregate itself (Target), or only to train or calibrate the method on other data, such as a submetered pilot (Training). Transferable: a calibration transfers to un-submetered aggregates.

Level: individual meter; summed meters of an aggregate; feeder, substation or transformer measurement; or a submetered population. AC: air conditioning; ADMD: after-diversity maximum demand; CNN: convolutional neural network; DER: distributed energy resource; DR: demand response; EV: electric vehicle; GHI: global horizontal irradiance; HVAC: heating, ventilation and air conditioning; ML: machine learning; NILM: non-intrusive load monitoring; PU: positive-unlabelled; PV: photovoltaics.

Methods infer the capacity from aggregated net load or smart-meter data [25], [26], [39], [40], and even recover the service-transformer rating this way [27]. Irradiance plays the role that temperature could play for TCLs, yet the thermostatic-load counterpart is largely missing [8].

The simultaneity of an aggregated TCL population, and the coincident peak it produces, inform network planning, as studied mostly for HPs. They set the design load [28], [29], [41] and bound the flexibility the population offers [32], [33], [42]. Closest to this paper, Anderson et al. [30] fit a probabilistic model of HP demand against temperature to submetered customer data, and use it to assess LV networks under a given HP penetration. Such models run in the forward direction, from a known number of HPs to the demand they cause. Planners measure these quantities on submetered populations of known size, rather than recover them from an un-submetered feeder.

A separate body of work models aggregated TCL populations for control. A population can be described through the thermal parameters of its devices, either to deliver load following [43] or as a stochastic battery whose power and energy limits bound its flexibility [21]. The state of such a population can be estimated from its aggregate power [20], and the air-conditioning demand of a feeder can be disaggregated in real time from feeder measurements [16]. These methods, however, assume known device models or learn them from building- and device-level data, and they work at control time scales. None of them addresses the planning question of how much

TCL capacity is installed behind the meter.

To the best of the authors' knowledge, no prior method estimates the installed capacity of an aggregated TCL population from feeder-level net load without submetering the target feeder. The change-point models observe the right signal but lack the step from coincident demand to installed capacity, while the simultaneity studies hold that step but only on submetered populations. The TCL aggregate models, in turn, assume the population is already known. This paper joins the first two. The contributions are threefold.

- 1) Estimation of the installed TCL capacity behind a feeder from its net load and temperature, without submetering the feeder. Established change-point models give the slope with which the daily net load rises below a balance-point temperature [38]. This paper adds the step from that coincident demand to installed capacity, by dividing it by the slope of the SF calibrated on a small set of submetered households. Each feeder keeps its own balance point, which follows its building stock and climate.
- 2) Evidence on when the SF slope transfers between groups of households. Within 50 submetered Swiss HPs, it is stable against the number of HPs and their capacity. On 1260 aggregates built from measured Swiss households, the capacity error is quantified against HP penetration, aggregate size and data resolution. On three real Swiss and German aggregates,

transferring the slope alone keeps the error within 10%, against up to 26% for the full SF curve. Across submetered households in five countries, the slope transfers within one climate zone, but not between climate zones or between social and private housing.

- 3) Estimation of the temperature-driven TCL energy and of a flexibility potential from the same two fits. The energy follows from the net-load slope alone, so it needs no submetered households. The flexibility is the energy the TCLs could shift within a window allowed by building thermal inertia, while delivering the same total energy. It equals the temperature-driven load times the unused share of capacity, one minus the SF, and is compared with the same quantity computed from hourly submetered data.

The rest of the paper is structured as follows. Section II states the problem, defines the net-load and SF fits and characterises them across datasets; Section III derives the energy, installed capacity and flexibility; Section IV evaluates them on semi-synthetic aggregates of one Swiss population; Section V applies the method to real Swiss and German aggregates outside the pilot; Section VI discusses accuracy, generalisation and data requirements. Lastly, Section VII concludes the paper.

II. PROBLEM AND APPROACH

Privacy concerns and limits on metering and data exchange leave DSOs without full observability of individual consumers. Their records of DERs also lag behind what is actually connected. DERs that consume or generate with a high simultaneity factor matter most for reliable operation, since they add to the peak together. This work therefore uses the load profiles a DSO does hold, aggregated in space at the feeder head or substation and in time over the day. From these, it estimates the installed capacity and flexibility potential of the thermostatic loads behind the meter.

A metered point of an LV network, such as a substation or feeder, aggregates the load of the consumers downstream. This paper calls such a metered point, together with the consumers behind it, an aggregate. Some of its consumers run TCLs. The task is to quantify those TCLs from the aggregate's net load.

The net load is the sum of a TCL and a non-TCL component,

$$P_{\text{net}}(t) = P_{\text{TCL}}(t) + P_{\text{non-TCL}}(t), \quad (1)$$

where only the TCL component responds directly to temperature. The non-TCL component also carries a weaker, indirect temperature dependence, visible in the whole-home energy signature [44]. Still, TCLs dominate the total temperature response, so the model neglects the non-TCL contribution. Section VI revisits this assumption.

The problem is to recover the aggregated TCL load $P_{\text{TCL}}(t)$ from the aggregated net load. From it follow three quantities that aid DSO operation and planning: the TCLs' installed capacity, their energy consumption and their flexibility potential. Fig. 1 sketches the pipeline.

Two fits describe the TCL population's response to ambient temperature. The first, the net-load fit, uses data the DSO already holds, with no submetering. The second, the SF fit, needs submetered TCL data. Both share the same piecewise-linear form and use daily means, which reduce the effect of building thermal inertia [18].

A. Net-Load Fit

When an aggregate's net load is averaged over each day and plotted against the daily mean temperature, it follows a piecewise-linear pattern. The load is roughly flat on mild days, and rises close to

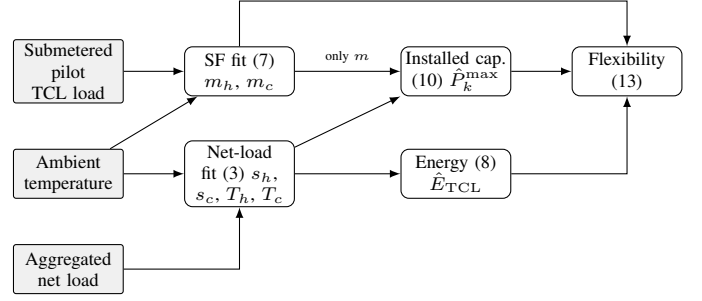


Fig. 1. Method pipeline. The net-load fit, which needs only data the DSO holds, supplies the sensitivities s_h , s_c and the thresholds T_h , T_c ; the SF fit of a submetered pilot supplies only the SF sensitivities m_h , m_c . Their ratio gives the installed capacity per regime. The net-load fit alone gives the temperature-driven energy, and the capacity with the transferred sensitivity gives the flexibility.

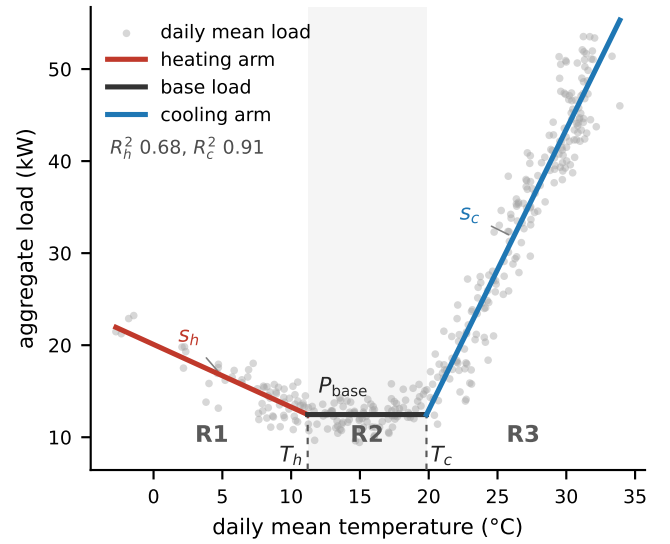


Fig. 2. Net-load fit (2) to the daily mean load of an aggregate of air-conditioned and furnace-heated households in Austin, Texas [45], with the heating (R1), dead-band (R2) and cooling (R3) regimes.

linearly as the temperature falls below, or rises above, a threshold. The net-load fit expresses the aggregated net load of (1) as

$$\hat{P}_{\text{net}}(T) = P_{\text{base}} + s_h \max(0, T_h - T) + s_c \max(0, T - T_c), \quad (2)$$

where P_{base} is the temperature-independent load [kW], s_h and s_c are the heating and cooling sensitivities [kW/°C], and $T_h \leq T_c$ are the heating and cooling threshold temperatures [°C], also known as balance-point temperatures, bounding a dead band over which no thermal response is present. The parameters of the fitted curve are computed through Ordinary Least Squares (OLS), with T_h and T_c being the values that minimise the squared residuals of the entire fit. The two temperature-dependent segments, called the heating and cooling arms of the fit, form the fitted aggregated TCL load.

Note that (2) needs only the net load and ambient temperature the DSO already holds, and no submetered or labelled ground truth. Fig. 2 shows the net-load fit to aggregated Austin smart-meter data [45], with the parameters of (2) marked. The thresholds divide the temperature axis into three regimes. R1 (below T_h) and R3 (above T_c) are the heating and cooling regimes, and R2 is the dead band between them, where the population neither heats nor cools. This work characterises R1 and R3 and treats R2 as out of scope.

For an aggregate i with its own fitted parameters,

$$\hat{P}_{\text{TCL}}^i(T) = s_h^i \max(0, T_h^i - T) + s_c^i \max(0, T - T_c^i), \quad (3)$$

where the hat marks a fitted function rather than a measurement. This estimated load only considers the temperature-driven component of TCL. The flat term P_{base} is the temperature-independent load, dominated by the non-TCL component of (1), and is discarded.

The fitted parameters carry a physical interpretation. The threshold temperatures T_h and T_c are behavioural. They mark where the population, on average, begins to heat or cool, rather than a fixed property of any building. Below T_h a heating load appears and above T_c a cooling load appears, each growing close to linearly with the distance from its threshold. Between the two lies a dead band where the population neither heats nor cools. Individual households start heating or cooling at different temperatures. Once many are aggregated, these differences average out, and T_h and T_c settle at values that reflect regional preferences for starting space heating or cooling (Section II-C).

The sensitivities s_h and s_c of the net-load fit measure how much heating or cooling load the population adds per degree. They scale with the TCL capacity installed behind the meter, so an aggregate with more TCLs shows a steeper arm, all else equal. Between aggregates of the same size, a steeper arm therefore also indicates a higher TCL penetration, i.e., a larger share of consumers with a TCL. Section IV-B confirms this, with s_h tracking installed capacity across the aggregates.

The flat centre P_{base} is the load that remains on days between T_h and T_c . The net load on these days is not constant, but it shows no trend with temperature. This temperature-independent load is outside the scope of this paper.

The balance between s_h and s_c reflects the local climate. Hot regions are expected to be cooling-dominated, with $s_h \approx 0$, whereas regions with long winters, such as the Nordic countries, are heating-dominated, with $s_c \approx 0$. Table II spans this range, from the heating-only Hamelin and Kloten aggregates to Maricopa, where cooling dominates.

B. Simultaneity-Factor Fit

The SF, also called the coincidence factor, is the ratio of the demand of a group of loads, here TCLs, at a given instant to the sum of their individual maximum demands. It reflects that connected devices do not all draw their full power at the same instant [46]. In distribution networks, the SF is central to planning. It lets utilities size transformers, cables and protection for the peak the group actually draws, rather than for the sum of individual peaks.

Mathematically, the SF is the fraction of the aggregate's installed capacity consumed at a given temperature T , and is given as

$$\text{SF}(T) = \frac{P_{\text{TCL}}(T)}{P_{\text{TCL}}^{\text{max}}}, \quad (4)$$

where $P_{\text{TCL}}(T)$ is the daily mean TCL load on a day of mean temperature T , and $P_{\text{TCL}}^{\text{max}}$ is the cumulative installed capacity of all the TCLs in the aggregate. In this work, the installed capacity of each TCL is taken as the 99.9th percentile of its submetered 15-min draw. The installed capacity of an aggregate is thus the summed peak electrical draw of its TCLs, which tracks their rated power while ignoring isolated metering spikes.

Since $P_{\text{TCL}}(T)$ is a daily mean, the SF of this paper is a daily-mean SF. It is lower than the coincidence factor used in network planning, which takes the peak interval. On the coldest day of the Kloten data, the 15-min peak SF of the 50 submetered HPs is 1.28 times the daily mean, and on a median heating day 1.75 times. The

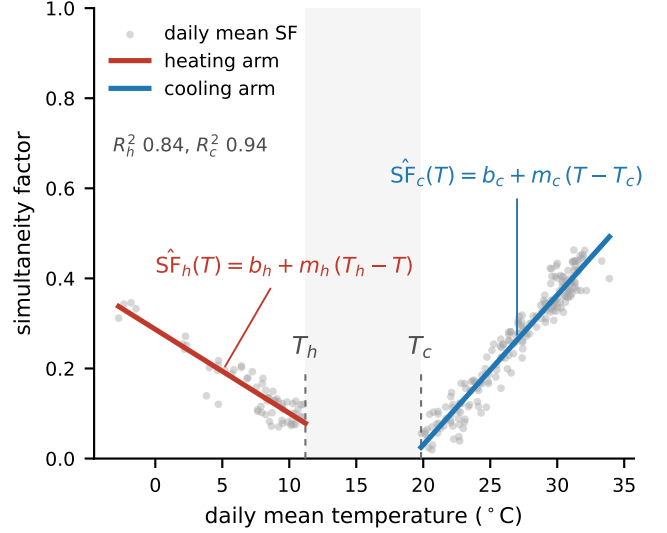


Fig. 3. SF fit (7) to the aggregated Austin TCLs [45], with each daily point normalised by the installed capacity of the TCLs active in its regime (6).

daily mean suits the aim of this paper, since the installed capacity follows from the slope of the SF, and daily means reduce the effect of building thermal inertia [18].

An aggregate may have a mix of TCLs with different behaviours with respect to temperature. While some TCLs can be active in different regimes of the net-load fit, such as reversible HPs that can heat and cool, others are exclusive to either heating (e.g. resistive heating) or cooling (e.g. air-conditioning). Using the total TCL installed capacity in (4) would lead to an underestimation of the SF, since the installed capacity of cooling-exclusive devices would still count towards the SF in the heating regime, and *vice-versa*. Therefore, each regime will have its own capacity, which is defined as

$$P_k^{\text{max}} = \sum_{n \in \mathcal{N}_k} P_n^{\text{max}}, \quad (5)$$

where k denotes the regime (heating or cooling), $\mathcal{N}_k$ is the set of TCLs active in regime k , and P_n^{max} is the installed capacity of TCL n . The true value of SF is then defined per heating and cooling regime as

$$\begin{aligned} \text{SF}_h(T) &= P_{\text{TCL}}(T)/P_h^{\text{max}}, \quad \forall T < T_h, \\ \text{SF}_c(T) &= P_{\text{TCL}}(T)/P_c^{\text{max}}, \quad \forall T > T_c, \end{aligned} \quad (6)$$

with h denoting heating and c cooling. It is by definition bounded in $[0, 1]$. The SF fit takes the same form as (2),

$$\begin{aligned} \hat{\text{SF}}_h(T) &= b_h + m_h(T_h - T), \quad \forall T < T_h, \\ \hat{\text{SF}}_c(T) &= b_c + m_c(T - T_c), \quad \forall T > T_c, \end{aligned} \quad (7)$$

where T_h and T_c are the thresholds of the net-load fit (2) of the same aggregate. The SF fit does not fit thresholds of its own. It borrows them from the net-load fit, so both fits describe the same heating and cooling regimes. The intercepts b_h and b_c and the heating and cooling SF sensitivities m_h and m_c [$1/^\circ\text{C}$] are determined by OLS on the days of each regime. The intercepts capture the weather-independent share of TCL use, such as domestic hot water, which the net-load fit instead places in P_{base} . Fig. 3 shows the SF data points and the SF fit for the same Austin population of Fig. 2.

The observed SF stays well below 1. Each arm would reach the installed capacity of its regime, with every TCL drawing at once, only at temperatures well outside any observed weather. The gap between

TABLE II
NET-LOAD AND SF FITS, ONE AGGREGATE PER DATASET

dataset	TCL tech	n	Climate	Net-load fit $\hat{P}_{\text{net}}(T)$							SF fit $\hat{S}\hat{F}(T)$							
				P_{base}	s_h	T_h	R_h^2	s_c	T_c	R_c^2	b_h	m_h	SF_h^{max}	R_h^2	b_c	m_c	SF_c^{max}	R_c^2
Hamelin, DE [47]	WSHP	37	Cfb	16	3.9	13.9	0.72	–	–	–	0.014	0.016	0.32	0.72	–	–	–	–
Kloten, CH [48], [49]	ASHP, GSHP	50	Cfb	54	6.1	16.5	0.92	–	–	–	0.029	0.018	0.47	0.95	–	–	–	–
Oslo, NO [50]	GSHP, ER	849	Dfb	516	52.5	15.2	0.88	–	–	–	0.106	0.022	0.70	0.80	–	–	–	–
Hennepin, MN, US [51] ^a	ASHP	150	Dfa	118	42.3	8.0	0.74	–	–	–	0.061	0.022	0.46	0.93	–	–	–	–
WA, US [52] ^b	DHP, ER	22	Csb	25	2.4	14.6	0.90	2.0	21.3	0.79	0.058	0.024	0.53	0.91	0.067	0.031	0.24	0.83
OR, US [52] ^b	DHP, ER	21	Csb	25	2.1	13.3	0.87	2.0	21.8	0.75	0.044	0.023	0.46	0.91	0.097	0.024	0.22	0.79
King, WA, US [51]	ASHP, ER	150	Csb	146	21.1	13.3	0.89	11.0	18.4	0.93	0.036	0.036	0.53	0.90	0.091	0.038	0.37	0.96
Austin, TX, US [45]	AC, ER	22	Cfa	12	0.7	11.2	0.68	3.0	19.8	0.91	0.078	0.019	0.34	0.84	0.025	0.033	0.49	0.94
Maricopa, AZ, US [51]	ASHP, ER	150	BWh	172	15.3	11.6	0.83	25.7	26.0	0.95	0.144	0.063	0.52	0.86	0.047	0.024	0.39	0.95

n : aggregated consumers. Climate: Köppen–Geiger class [53]: Cfb, temperate oceanic; Dfa, Dfb, humid continental; Csb, warm-summer Mediterranean; Cfa, humid subtropical; BWh, hot desert. R_h^2 , R_c^2 : fit scores in R1 and R3. ASHP, GSHP, WSHP: air-, ground-, water-source HP; DHP: ductless (mini-split) HP; AC: air conditioning; ER: electric resistance heating.

^a Hennepin’s SF uses the ASHP compressor’s own draw only, excluding its electric-resistance HP-backup element

^b The SF uses the submetered ductless HPs only. Most of these homes also have electric baseboard, furnace or other zonal heating circuits (17 of 22 in WA, 16 of 21 in OR), which the net load includes.

the observed TCL load and the installed capacity, the unused share $1 - \text{SF}$ of that capacity, is what the flexibility of Section III-C draws on.

C. Cross-dataset characterisation

Table II reports both fits, $\hat{P}_{\text{net}}(T)$ and $\hat{S}\hat{F}(T)$, on aggregates from four countries and a range of climates, with each climate given as its Köppen–Geiger class [53]. R1, R3, or both are resolved as the local climate dictates, and each is scored on the days of its own regime. R1 fits with a net-load R_h^2 between 0.68 and 0.92 across the year-round aggregates, lowest in Austin where electric heating is marginal. R3 ranges from R_c^2 of 0.95 in hot Maricopa to 0.75 in mild Oregon, where cooling is rare.

The threshold temperatures follow the local climate. T_h falls between 8 and 17 °C, and T_c between 18 and 26 °C. A hot county such as Maricopa has a steep R3, whereas a cold one such as Hamelin has none. The sensitivities s_h and s_c of the net-load fit scale with the aggregate’s total installed TCL, and therefore with its size. The 849-household Norwegian aggregate and the electrified ResStock counties show the steepest arms for this reason. The SF sensitivities m_h and m_c remove this size dependence.

The SF sensitivities m_h and m_c group partly by Köppen class. The two temperate oceanic (Cfb) aggregates, Hamelin and Kloten, sit at $m_h = 0.016$ and 0.018 °C^{−1}, despite running on water-source and mixed ground-and-air-source heat pumps respectively. The two humid continental (Dfa and Dfb) sites, Oslo and Hennepin, converge on $m_h \approx 0.022$ °C^{−1}, even though Oslo’s heating load is mostly ground-source and direct resistance, while Hennepin’s is air-source. The temperate dry and warm summer (Csb) class, however, splits. Its two measured aggregates, the Washington and Oregon households with ductless HPs, sit at $m_h = 0.023$ – 0.024 °C^{−1}, whereas the simulated King County aggregate of ducted HPs with resistance backup reaches 0.036 °C^{−1}. The cooling sensitivity m_c spans 0.024 – 0.038 °C^{−1} without a clear grouping. The hot desert site, Maricopa, is the outlier, with the steepest heating sensitivity of all, $m_h = 0.063$ °C^{−1}. This is possibly because a single aggregate and few genuinely cold days leave little data to anchor the coldest end of its heating arm. Overall, the Köppen class narrows m_h for aggregates of similar TCL technology, but the technology, and possibly the difference between measured and simulated peaks, matters as well. Across classes, m_h varies by up to a factor of four, so the transfer of Section II-D is only expected to hold within a climate class and TCL technology. Section V-B tests this expectation on further populations with complete heating submetering.

D. Transferability of SF

The SF is computed from submetered TCL data, which a DSO rarely holds for the aggregate under study. The SF fit (7) is therefore calibrated on a submetered pilot and reused on aggregates of the same population that have no submetering. Throughout, a population is the set of households of one dataset or study; its pilot is the submetered part used for calibration, a target is an aggregate the estimate is applied to, and a split is one random partition of a population into a pilot and targets. Only the SF sensitivities m_h and m_c transfer. The pilot’s thresholds come from the pilot’s own net-load fit and follow its building stock and climate, so the target’s thresholds T_h and T_c are read from the target’s net-load fit instead. The pilot’s intercepts b_h and b_c describe weather-independent TCL use, which the target’s net-load fit places in P_{base} and never attributes to its TCL arms.

The transfer rests on one hypothesis: within a population, m_h and m_c do not track the installed capacity or the TCL penetration of the aggregate. Section IV-C tests this hypothesis within the Kloten population, and Section V tests it across populations. If it holds, the transferred sensitivity turns the net-load fit into an estimate of the unobserved installed capacity, through (10). This is the mechanism the rest of the paper builds on.

III. AGGREGATED TCL QUANTIFICATION

The two fits, (3) and (7), are complementary. The net-load fit (3) needs no submetering, but its sensitivities also absorb any temperature response of the non-TCL load. The SF fit (7) needs submetering but isolates the TCL simultaneity. Together they estimate three quantities a DSO needs about the TCLs of an aggregate: their energy consumption, installed capacity and flexibility potential. Energy consumption follows from (3) alone. Installed capacity and flexibility also need the transferred SF sensitivity, since the net load exposes the aggregated TCL demand, not the non-coincident sum of the individual peaks. Section IV evaluates each quantity.

A. TCL Energy Consumption Estimation

The energy the TCL population draws follows from integrating its temperature response over a certain period. Over a day of mean temperature T , the temperature-driven TCL energy is given by

$$\hat{E}_{\text{TCL}} = (s_h \max(0, T_h - T) + s_c \max(0, T - T_c)) \Delta t, \quad (8)$$

with Δt being the length of the period, here one day. The annual energy is the sum of (8) over the days of the year. The fitted TCL load alone suffices. Given a measured temperature, it recovers the

aggregated TCL load $\hat{P}_{\text{TCL}}(T)$ from the net load $P_{\text{net}}(t)$; given a forecast temperature, it forecasts that load for a future period.

R2 bounds the estimate. Between T_h and T_c both arms are zero. The estimate therefore misses any TCL energy that persists into mild weather, such as domestic hot water. The estimate captures the temperature-driven share, not the annual total.

B. TCL Installed Capacity Estimation

The installed capacity is the sum of the individual peak demands (5), defined per regime. The aggregated TCLs never draw it at once, so the net load never exposes it directly. The net-load fit does expose the temperature-driven TCL load $\hat{P}_{\text{TCL}}^i(T)$ of aggregate i . Rearranging (6), the capacity of regime k follows from this load and the SF at the same temperature.

Both must describe the same load. The net-load arm is zero at the aggregate's own threshold by construction, and the weather-independent TCL use sits in P_{base} . The matching SF is therefore the temperature-driven part of (7). It starts from zero at the aggregate's own thresholds and grows with the transferred sensitivity,

$$\begin{aligned}\hat{\text{SF}}_h^i(T) &= m_h (T_h^i - T), \quad \forall T < T_h^i, \\ \hat{\text{SF}}_c^i(T) &= m_c (T - T_c^i), \quad \forall T > T_c^i.\end{aligned}\quad (9)$$

Dividing the arms of (3) by (9) gives the installed capacity of each regime,

$$\begin{aligned}\hat{P}_h^{\text{max},i} &= \frac{s_h^i (T_h^i - T)}{m_h (T_h^i - T)} = \frac{s_h^i}{m_h}, \quad \forall T < T_h^i, \\ \hat{P}_c^{\text{max},i} &= \frac{s_c^i (T - T_c^i)}{m_c (T - T_c^i)} = \frac{s_c^i}{m_c}, \quad \forall T > T_c^i.\end{aligned}\quad (10)$$

The temperature cancels, so the estimate needs no choice of evaluation day. The net-load fit supplies the numerator, which the DSO observes. The pilot supplies only the denominator m_k . The estimate is therefore only as good as the transferred sensitivity, which Section IV tests.

The two regimes are estimated separately. Their sum equals the total installed TCL capacity only when no device serves both regimes. A reversible HP counts in both $\hat{P}_h^{\text{max}}$ and $\hat{P}_c^{\text{max}}$. For a population of reversible HPs, the total therefore lies between the larger of the two and their sum.

C. Flexibility Quantification

The flexibility a TCL population offers is the energy it can shift from its baseline consumption without breaching indoor thermal comfort [42]. Building-stock insulation and indoor-comfort data are lacking, so an abstract time window Δt_{Th} stands in. Within it, the population must deliver a fixed thermal energy to stay comfortable. A larger Δt_{Th} means a better-insulated building, or wider comfort bands [54]. Over this Δt_{Th} , the aggregated TCLs must deliver the temperature-set energy demand. This is the temperature-driven TCL energy of (8), evaluated over the window Δt_{Th} and drawn at the baseline power $\hat{P}_{\text{TCL}}(T) = \hat{\text{SF}}(T) \hat{P}^{\text{max}}$,

$$\hat{E}_{\text{TCL}}^{\Delta t_{Th}} = \hat{P}_{\text{TCL}}(T) \Delta t_{Th} = \hat{\text{SF}}(T) \hat{P}^{\text{max}} \Delta t_{Th}, \quad (11)$$

where $\hat{\text{SF}}$ and $\hat{P}^{\text{max}}$ are those of the active regime, from (9) and (10). Their product equals the fitted TCL load of (3), so the baseline is the temperature-driven load of the net-load fit. The schedule that delivers this same energy while deviating most from the baseline is the two-level square wave of Fig. 4: full installed power $\hat{P}^{\text{max}}$ for a fraction α_F of the window and zero for the rest. Its energy $\alpha_F \hat{P}^{\text{max}} \Delta t_{Th}$ must equal (11), which fixes the duty cycle at the simultaneity factor,

$$\alpha_F \hat{P}^{\text{max}} \Delta t_{Th} = \hat{\text{SF}}(T) \hat{P}^{\text{max}} \Delta t_{Th} \implies \alpha_F = \hat{\text{SF}}(T). \quad (12)$$

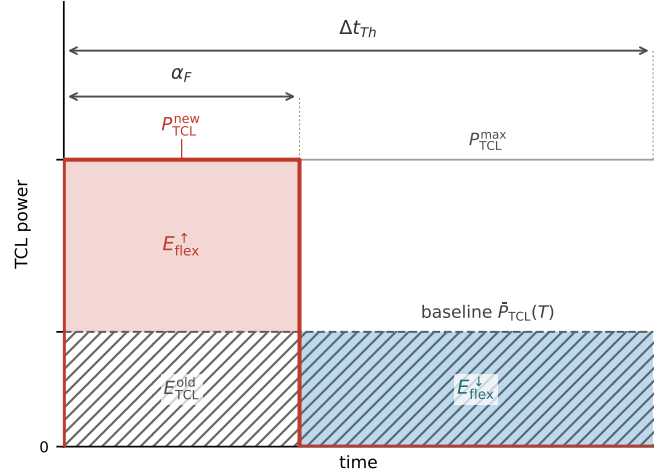


Fig. 4. The flexibility of (13) over one cycle of the square-wave schedule. The aggregate must draw the baseline energy $E_{\text{TCL}}^{\text{old}} = \hat{P}_{\text{TCL}}(T) \Delta t_{Th}$ (hatched) over the window. It can instead draw the same energy as the full-power pulse $P_{\text{TCL}}^{\text{new}}$ of duty cycle $\alpha_F = \hat{\text{SF}}(T)$. The deviation from baseline is the flexible energy, equal upward ($E_{\text{flex}}^{\uparrow}$) and downward ($E_{\text{flex}}^{\downarrow}$).

The flexible energy is this schedule's symmetric deviation from the baseline. During the on phase the population draws $\hat{P}^{\text{max}} - \hat{P}_{\text{TCL}}$ above baseline for a time $\alpha_F \Delta t_{Th}$; during the off phase it curtails $\hat{P}_{\text{TCL}}$ for a time $(1 - \alpha_F) \Delta t_{Th}$. The two areas are equal, giving an upward and downward flexible energy

$$E_{\text{flex}} = \hat{\text{SF}}(T) (1 - \hat{\text{SF}}(T)) \hat{P}^{\text{max}} \Delta t_{Th}. \quad (13)$$

This use case draws on both fits and one external parameter. Since $\hat{\text{SF}}(T) \hat{P}^{\text{max}}$ equals the fitted TCL load, (13) equals $\hat{P}_{\text{TCL}}(T) (1 - \hat{\text{SF}}(T)) \Delta t_{Th}$. The net-load fit sets the load that can be shifted, and the transferred sensitivity only the unused capacity above it, so the capacity estimate cancels. The window Δt_{Th} , over which the population can be reshaped, is a property of the building stock, not of the fits. The instantaneous unused capacity $(1 - \hat{\text{SF}}(T)) \hat{P}^{\text{max}}$ is the special case of the on phase drawn to full power.

Over a year, this paper counts one such window per day of R1 or R3, the regimes in which (9) is defined. The annual flexible energy is then the sum of (13) over these days, each evaluated at its daily mean temperature, and it scales linearly with Δt_{Th} . The days of R2 are left out of the estimate and of its ground truth alike. The estimate therefore covers only temperature-driven flexibility; weather-independent use that could also be shifted, such as hot-water heating, falls outside it.

The bound is an idealisation, not a dispatchable schedule. It takes the square wave as feasible and ignores each building's thermal state and the comfort limits within which its TCLs can shift. It therefore stands as a potential against temperature and window length, not as a firm reserve.

IV. NUMERICAL RESULTS

A. Kloten Data and Evaluation Design

This section uses a single Swiss population, the households metered at the Kloten weather station in calendar 2023. The data combines two sources, the HEAPO dataset [48] and an additional smart-meter dataset [49]. The load profiles have 15-minute resolution, while the ambient temperature is hourly. The only TCL used throughout is the HP and only heating is observed. The station holds 50 households

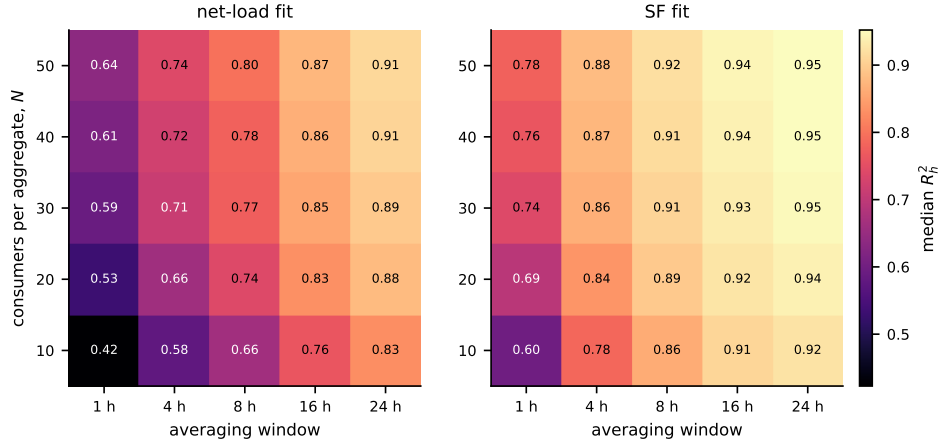


Fig. 5. Median R_h^2 of the net-load fit (left) and the SF fit (right) of the Kloten aggregates against spatial aggregation (consumers per aggregate, N) and temporal aggregation (averaging window), at a fixed 50% HP penetration.

with a submetered HP and 1307 households with neither an HP nor any other electric heating.

The 50 HPs are split at random into two halves of 25, twenty times over. In each split, the first half is the submetered pilot: its aggregated SF is fitted with (7) and supplies the sensitivity m_h . Semi-synthetic aggregates, i.e. sums of measured households, are then drawn from the second half only, so no HP informs both the pilot and a target. Each aggregate combines N_{hp} distinct HPs of that half, with the same households' other load, and $N - N_{hp}$ distinct households without an HP. The consumer count N takes the values 10, 20, 30, 40 and 50, and N_{hp} the values 5, 10, 15, 20 and 25, with $N_{hp} \leq N$. Every feasible cell holds three aggregates per split, giving 1260 aggregates. All draws are without replacement, so no household recurs within an aggregate.

Two error metrics recur below. The weighted absolute percentage error (WAPE) scores an estimate $\hat{y}_i$ against its ground truth y_i across the aggregates i ,

$$\text{WAPE} = \frac{\sum_i |\hat{y}_i - y_i|}{\sum_i y_i} \times 100\%, \quad (14)$$

so WAPE weights each aggregate by its true magnitude, not equally. A small aggregate, with a large percentage error but a small absolute error, therefore cannot dominate. The metric reports the total error across a set of aggregates, as a DSO would read it. The median ratio, the median of $\hat{y}_i/y_i$, gives the direction of the error, < 1 for an underestimate and > 1 for an overestimate.

B. Fit Quality across Spatial and Temporal Aggregation

Both fits, (2) and (7), are made to each aggregate's measurements, varying the number of consumers and the resolution. The HP penetration is kept fixed at 50%, on a separate set of Kloten aggregates, also drawn without replacement. Spatial aggregation averages out the intermittent, non-thermal behaviour, as more meters are summed behind one aggregate. Temporal aggregation does the same over time, averaging the load over longer windows removes the occupancy swings and building thermal inertia that do not track temperature one-to-one at finer resolutions. Fig. 5 reports both, as the median R_h^2 of the net-load and SF fits over every consumer-count/averaging-window cell at this fixed HP penetration. Both fits are scored on the points below the net-load threshold T_h , at which the SF fit is anchored.

Each column shows the spatial effect, read from ten consumers at the bottom of the grid to fifty at the top. In the hourly column the

net-load fit rises from a median R_h^2 of 0.42 at ten consumers to 0.64 at fifty, and the SF fit rises the same way, from 0.60 to 0.78. The same rise holds in the daily column, where the net-load fit climbs from 0.83 to 0.91 and the SF fit from 0.92 to 0.95.

Each row shows the temporal effect, read from hourly resolution on the left to daily on the right. At ten consumers the net-load fit climbs monotonically from a median R_h^2 of 0.42 at hourly resolution through 0.58 and 0.66 at four and eight hours, to 0.76 at sixteen and 0.83 at daily; the SF fit climbs the same way, from 0.60 through 0.78 and 0.86 to 0.91 and 0.92. The same climb holds in every other row: at fifty consumers, for instance, the net-load fit reaches 0.87 and 0.91 in the two rightmost cells and the SF fit reaches 0.94 and 0.95. The daily mean, used throughout the rest of the paper, therefore sits at the right edge of this curve, with sixteen hours already close behind it.

The two effects compound along the grid's diagonal. The weakest cell, ten consumers at hourly resolution, gives a median R_h^2 of 0.42 for the net-load fit and 0.60 for the SF fit; the strongest, fifty consumers at daily resolution, reaches 0.91 and 0.95. This suggests the two forms of aggregation are, to some extent, substitutable: a coarser averaging window recovers much of what a larger aggregate would, and a larger aggregate recovers much of what a coarser window would.

The heating sensitivity itself scales with the population behind the meter. Fig. 6 plots the fitted s_h against the installed HP capacity across the aggregates, coloured by the number of HPs. The two track closely, at a Pearson correlation of 0.97, along a line through the origin of slope 0.020°C^{-1} . A steeper R1 arm therefore signals more installed capacity, as Section II-A anticipated. This slope exceeds the SF sensitivity of Section IV-C by about 12%, a first sign of the non-TCL temperature response that the net-load sensitivity absorbs.

C. Simultaneity Factor: Invariance and Transfer

Beyond fitting well, the SF fit is stable. Because the SF comes from the submetered HPs alone, the households without an HP do not affect it. Table III therefore reports the aggregates of a constant size, 50 consumers, where N_{hp} is the only variable. Neither the maximum heating SF nor the sensitivity m_h changes meaningfully with the number of HPs or their installed capacity: m_h holds near 0.018°C^{-1} and the maximum SF near 0.48. More HPs behind the meter do not shift these values but tighten them, as the interquartile range of m_h narrows from 0.0047 at five HPs to 0.0013 at twenty-five. This

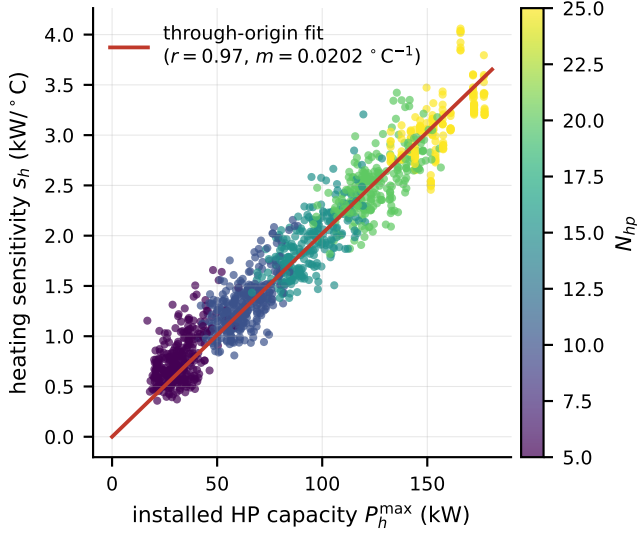


Fig. 6. Fitted heating sensitivity s_h against installed HP capacity across the 1260 Kloten aggregates, coloured by the number of HPs.

TABLE III
SF FITS OF THE KLOTEN AGGREGATES, MEDIAN (INTERQUARTILE RANGE, IQR)

N_{hp}	R_h^2	$SF_h^{\max}$	b_h	m_h ($^{\circ}\text{C}^{-1}$)
5	0.92 (0.04)	0.48 (0.11)	0.022 (0.028)	0.0182 (0.0047)
10	0.94 (0.02)	0.47 (0.08)	0.025 (0.016)	0.0177 (0.0036)
15	0.95 (0.02)	0.47 (0.06)	0.027 (0.010)	0.0177 (0.0023)
20	0.95 (0.01)	0.48 (0.05)	0.026 (0.009)	0.0183 (0.0018)
25	0.95 (0.01)	0.48 (0.03)	0.026 (0.007)	0.0180 (0.0013)

maximum is a daily mean. Taken at the 15-min peak, as network planning takes it, the SF of the 50 Kloten HPs reaches 0.56 on the coldest day and 0.59 over the year, above the HP SF of 0.425 that national LV network studies draw from field trials [55].

The pilot transfers the SF to the aggregates without submetering. Fig. 7 shows the pilot of one split, whose SF fit, at the pilot's own net-load threshold, gives $\hat{SF}_h(T) = 0.021 + 0.019(16.6 - T)$ for $T < 16.6^{\circ}\text{C}$, at an R_h^2 of 0.94. Across the twenty splits, the pilot sensitivity m_h has a median of $0.018^{\circ}\text{C}^{-1}$ and an interquartile range of 0.017–0.019. This sensitivity is the only parameter that (10) and the flexibility estimate transfer to a target.

A direct test sets the pilot against the whole other half, over 120 random splits. Placed at the other half's net-load threshold, the pilot's SF curve reproduces the SF of that half at a median R_h^2 of 0.91, with an interquartile range of 0.85–0.94. The pilot's sensitivity also recovers the capacity of the other half through (10). The median error is +8%, with an interquartile range from 0 to +18%. Section IV-E uses the pilot sensitivity to estimate installed capacity across the aggregates.

D. Energy Consumption Estimation

Equation (8) needs only the net-load fit, so the energy is the one estimate that requires no pilot. On the days below the threshold, R1 is compared against the submetered HP energy of each aggregate. Across the 1260 aggregates, the estimate reaches a WAPE of 9% at a median ratio of 0.97. About 7% of the annual HP energy falls in R2, most plausibly domestic hot water [11], which (8) omits by construction, so the annual ratio falls to 0.90. Integrating the daily-fitted arm over the hourly temperature, rather than the daily mean,

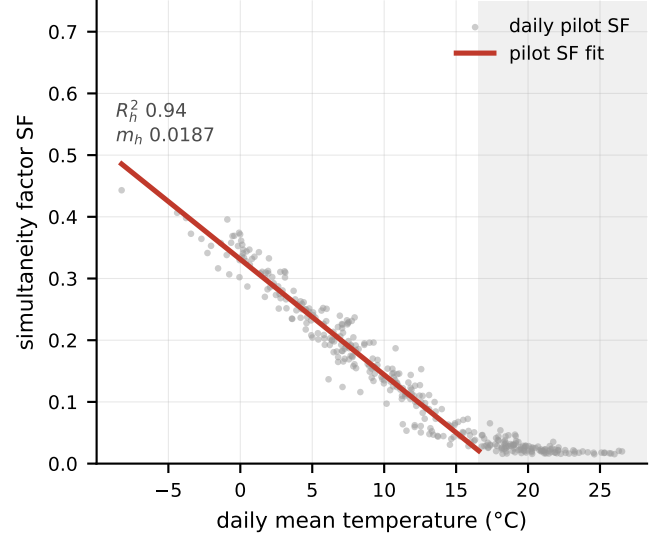


Fig. 7. SF fit to the aggregated pilot half of the Kloten HPs in one random split (grey), at an R_h^2 of 0.94. Its sensitivity m_h is the parameter transferred to the target aggregates.

recovers part of this, lifting the annual ratio to 0.94, since a day that crosses the threshold holds more heating-degree-hours than its mean implies. The same R1 energy is the baseline of the flexibility in Section IV-F.

E. Installed Capacity Estimation

Equation (10) estimates the installed HP capacity of each aggregate as the ratio of two sensitivities. The numerator s_h is read from the aggregate's own net-load fit. The denominator m_h is the sensitivity of the pilot of the same split (Section IV-C). The target aggregate is therefore never submetered.

With the transferred sensitivity, (10) recovers the installed capacity of the 1260 aggregates at a WAPE of 19%, with a median overestimate of 16%. In absolute terms, the mean error is 17 kW, the capacity of about three HPs of the mean 6.3 kW. The overestimate is consistent with the non-TCL temperature response that the net-load sensitivity absorbs, the neglected term of (1) discussed in Section VI. The error shrinks as HPs are added: the WAPE falls from 41% at five HPs to 15% at twenty-five, and the median overestimate from 35% to 12%. This is possibly because the households without an HP, whose temperature response inflates the heating arm, weigh less against more HPs.

Fig. 8 repeats the estimate over the aggregation grid of Section IV-B, at a fixed 50% HP penetration. In each cell, both the target's net-load fit and the pilot's SF fit are made at the cell's resolution. The averaging window barely matters: along each row, the WAPE varies by at most 1.5 points, as both sensitivities of (10) come from fits at the same resolution. The consumer count matters more. The WAPE falls from 19% at ten consumers to between 12% and 14% at fifty, while the median overestimate stays between 9% and 14%. Five HPs among ten consumers thus give a WAPE of 19%, against 41% for five HPs across all consumer counts, which points to the HP penetration rather than the HP count. Unlike the fit quality of Fig. 5, the capacity estimate needs no resolution finer than the daily mean.

Table IV sets the estimate against three alternatives on the same aggregates. Dividing each aggregate's net-load sensitivity by its own SF sensitivity, rather than by the pilot's, removes the transfer, so

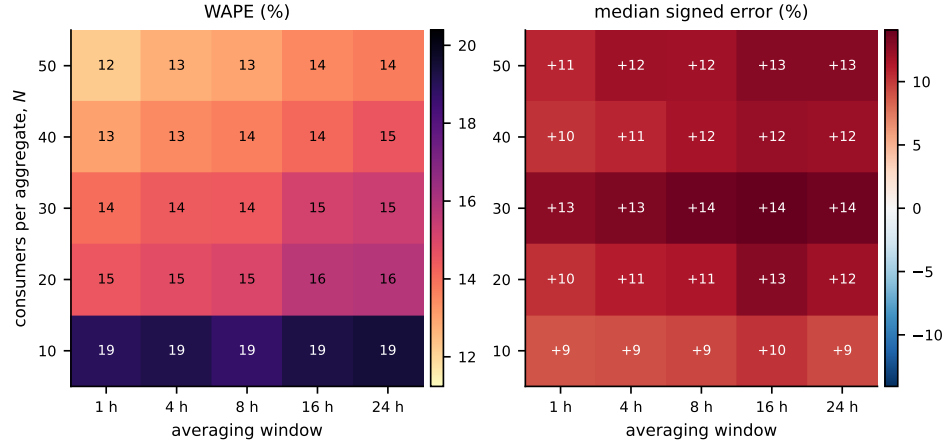


Fig. 8. Capacity error of (10) on the Kloten aggregates against spatial aggregation (consumers per aggregate, N) and temporal aggregation (averaging window), at a fixed 50% HP penetration. Left: WAPE; right: median signed error.

TABLE IV
CAPACITY WAPE (%) ON THE KLOTEN AGGREGATES, BY HP
PENETRATION N_{hp}/N

estimate	all	$\leq 15\%$	15–35%	35–65%	$> 65\%$
s_h/m_h , pilot m_h (10)	19.3	57.5	26.3	17.0	14.2
s_h/m_h , target m_h	14.5	53.0	22.5	12.6	8.1
full pilot SF curve	15.4	50.0	20.9	13.1	11.4
planning SF, $\hat{P}_{TCL}(T_{\min})/0.425$	21.1	64.6	28.8	19.0	14.5

the remaining error comes from the net-load sensitivity alone. This estimate reaches a WAPE of 14.5%, against 19.3% with the pilot's sensitivity: the transfer adds about five points, and the non-TCL temperature response in the net-load sensitivity accounts for the rest. Every estimate degrades in the same way as the HP penetration falls, from 8–15% above 65% penetration to 50–65% below 15%. This points to the net-load sensitivity as the dominant error at low penetration.

Transferring the full SF curve of the pilot, intercept and threshold included, and evaluating it on the coldest day performs better than the sensitivity alone, at a WAPE of 15.4%. This is possibly because the pilot's intercept, $b_h \approx 0.03$, raises the denominator and so partly offsets the upward bias of the net-load sensitivity. The offset relies on the pilot and the target sharing a threshold temperature, as all Kloten aggregates do, whereas the sensitivity alone needs no such match. A planning estimate divides the coldest-day TCL load of the net-load fit by the HP SF of 0.425 used in national LV studies [55]. It reaches a WAPE of 21.1%, since the Kloten span of 25 °C between threshold and coldest day happens to bring the coldest-day SF close to 0.425. Its error therefore depends on how cold the observed year is, whereas the proposed estimate carries no such dependence, since the temperature cancels in (10).

F. Flexibility Potential

The capacity estimate enters (13) only in appearance. Since $\hat{SF}(T) \hat{P}^{\max}$ equals the fitted TCL load, the estimate reduces to $\hat{P}_{TCL}(T) (1 - \hat{SF}(T)) \Delta t_{Th}$: the temperature-driven load of the net-load fit, times the unused share of capacity, $1 - \hat{SF}(T)$, left by the transferred SF. The actual flexibility is computed at hourly resolution: each hour's submetered SF gives $SF(1 - SF) P_h^{\max}$ with the submetered capacity, averaged over the day. Both are summed over the days of R1, below the aggregate's net-load threshold, since

the Kloten aggregates have no cooling regime. The estimate and its ground truth share no input other than the HP load contained in the net load, so their agreement does not follow by construction.

Fig. 9 gives the distribution of the estimated-to-actual ratio of annual flexible energy across the 1260 aggregates. The ratio has a median of 1.04 and an interquartile range of [0.97, 1.17], at a WAPE of 10%. It splits into two factors. The R1 energy of Section IV-D contributes a median ratio of 0.97, and the unused capacity a median ratio of 1.07. About three points of this unused-capacity factor come from the transferred SF running slightly below the measured one, and about four from the daily resolution of the estimate, since the SF swings within the day. The flexibility and energy ratios correlate at 0.96. The flexibility estimate therefore inherits its accuracy mainly from the net-load fit, while the transferred sensitivity shifts the unused capacity by a few per cent. Since Δt_{Th} cancels in the ratio, neither factor depends on the window.

Computed from the daily mean SF instead, as the estimate is, the actual flexibility would be 3% higher and the median ratio 1.01, since $SF(1 - SF)$ is concave. The days of R2, left out by construction, would add a weather-independent floor, most plausibly hot-water heating, worth a median 9% of the actual flexible energy over the whole year. On the coldest day, the estimate recovers the flexible energy at a WAPE of 21% and an R^2 of 0.73 against the actual values across aggregates.

Fig. 10 shows the daily actual and estimated flexible energy over the year for the held-out half of one split: 25 HPs with $P_h^{\max} = 144$ kW, drawn for an illustrative window $\Delta t_{Th} = 4$ h. None of these HPs informs the pilot, so the example is out of sample. The potential follows the season and is largest in winter, when the thermal demand that can be shifted is largest. The estimate tracks this profile day to day. A third of the days lie above the threshold, mostly from late spring to early autumn, and fall outside the count. On them, the measured flexibility keeps a floor of about 19 kWh per day, near 14% of its winter maximum, which the estimate omits by construction. On the coldest days the $SF(1 - SF)$ product nears its 0.25 ceiling, a quarter of $P_h^{\max} \Delta t_{Th}$, in both the actual and the estimate. Over the R1 days of the year, the flexible energy rises by 5.2 MWh per hour of window for the actual and 4.7 MWh for the estimate.

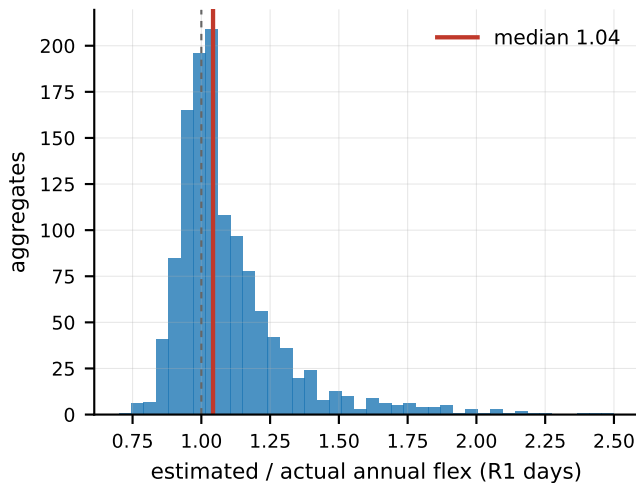


Fig. 9. Distribution across the 1260 Kloten aggregates of the estimated-to-actual ratio of annual flexible energy on the R1 days, with the median marked. The estimate uses the net-load fit and the transferred sensitivity; the actual uses each aggregate's own hourly submetered SF and submetered capacity.

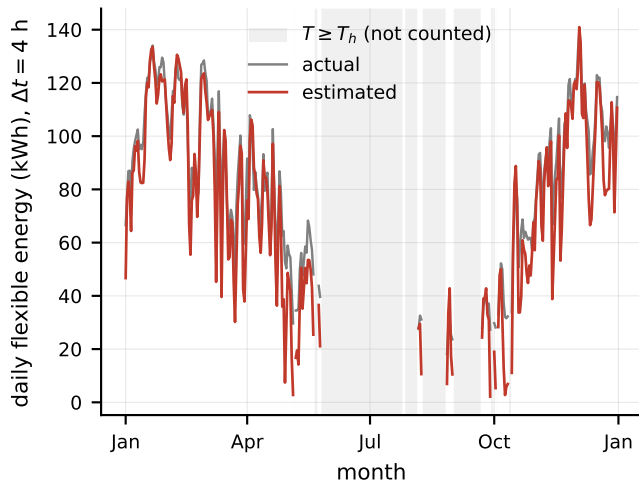


Fig. 10. Daily flexible energy of a held-out half of the Kloten HPs (25 HPs, $P_h^{\max} = 144$ kW) over the year, for an illustrative window $\Delta t_{T_h} = 4$ h; since E_{flex} is linear in Δt_{T_h} , the window sets only the vertical scale. Estimated from the pilot sensitivity of the same split versus computed from the hourly submetered SF; the shaded days of R2 are not counted.

V. REAL AGGREGATES

A. Real Aggregates outside the Pilot

Section IV tested the method within one population, on aggregates built from its households. This section applies it to real aggregates that no pilot has seen, each summed from its own field meters. Swiss and German aggregates test the heating regime with the Kloten pilot. Submetered data serve only to validate the results, never as input.

Three real aggregates lie outside the Kloten pilot. Two are the HEAPO households of two further Swiss weather stations, Hg and MqO, with 21 and 6 submetered HPs. The third comes from the WPuQ measurement campaign in Hamelin [47]: its net load is the sum of 37 individually metered German single-family houses, each with a submetered HP. All three are summed rather than metered at a transformer, so they carry no network losses. They also contain no households without an HP, which makes their HP penetration 100%.

TABLE V
TRANSFER TO REAL AGGREGATES OUTSIDE KLOTEN

aggregate	N_{hp}	$P_h^{\max}$ (kW)	$\hat{P}_h^{\max}$ (kW)	capacity error (%)				
				pilot m_h	target m_h	full curve	planning SF	flex
Hg, CH	21	109	112	+2	+6	-3	+4	0.96
MqO, CH	6	51	48	-6	+5	-20	-6	0.93
Hamelin, DE	37	239	217	-9	+3	-26	-26	0.93

The pilot now comprises all 50 Kloten HPs. Its SF fit, at the pilot's own net-load threshold, gives $\hat{SF}_h(T) = 0.029 + 0.018(16.6 - T)$, and only its sensitivity $m_h = 0.018^\circ\text{C}^{-1}$ transfers.

Table V reports the three estimates. The capacity errors are +2% at Hg, -6% at MqO and -9% at Hamelin. MqO holds only six HPs, below the ten from which Section IV-E finds the estimates reliable, so its result is indicative. The flexibility ratio on the R1 days lies between 0.93 and 0.96, slightly below the Kloten median of 1.04 (Section IV-F). It follows the R1 energy ratios of 0.89 to 0.92, while the unused-capacity factor stays between 1.01 and 1.05. The energy ratios are lower than on Kloten possibly because these aggregates hold no households without an HP, whose temperature response inflates the heating arm.

The thresholds show why only the sensitivity transfers. The net-load thresholds are 16.7°C at Hg, 14.0°C at MqO and 13.9°C at Hamelin, against 16.6°C for the Kloten pilot. The threshold temperature varies with the insulation of the building stock, the heat released indoors by occupants and appliances, and the preferred indoor setpoint [56]. Transferring the full SF curve from Kloten, intercept and threshold included, gives capacity errors of -3%, -20% and -26%. The error of the full curve therefore grows with the distance between the target's threshold and the pilot's, whereas the sensitivity alone stays within 10%. Unlike within Kloten, the full curve loses to the sensitivity alone here, since the targets' thresholds lie up to 2.7°C below the pilot's. The planning estimate of Section IV-E errs by +4%, -6% and -26%: at Hamelin the span between threshold and coldest day is only 19°C , against 25°C at Kloten, so the coldest-day SF falls well below 0.425. With each target's own SF sensitivity, the errors are +6%, +5% and +3%. This error of the net-load sensitivity remains small since every household in these aggregates has an HP.

B. Transfer across Populations

The three real aggregates test the full estimate, but from one Swiss pilot only. Table VI widens the test of the SF transfer to every population with submetered heating available to this study: 849 Oslo apartments grouped in ten blocks [50], 182 households of the GB RHPP trial [28], [57], and US households of the BPA [58] and NREL [59] field studies, whose submetering covers the HP and its backup heat. The BPA households lie in two heating zones, the milder western Washington (HZ1) and the colder inland (HZ2), where the NREL households also lie. On these datasets, only the SF transfer is tested. The RHPP, BPA and NREL data hold the heating load of each household but no other load, so no faithful net load can be built from them. In Oslo, the apartments' own load includes electric heating that the submetered heating channel omits. Each row of Table VI therefore compares the target's own SF sensitivity with the pilot's, and the ratio minus one is the capacity error caused by the transfer alone.

Between random splits of one population, the ratio has a median of 1.00 in every dataset. Its spread narrows as the pilot and target grow, from an interquartile range of about $\pm 20\%$ at ten RHPP households

TABLE VI
RATIO OF THE TARGET'S OWN TO THE PILOT'S SF SENSITIVITY [IQR
OVER RANDOM SPLITS OR A BOOTSTRAP OVER HOMES]

pilot → target		homes	$m_h^{\text{own}}/m_h^{\text{pilot}}$
<i>Random splits</i>			
Kloten → Kloten	CH	25/25	1.00 [0.92, 1.08]
Oslo → Oslo	NO	5/5 blocks	1.00 [0.91, 1.10]
RHPP → RHPP	GB	10/10	1.00 [0.83, 1.20]
RHPP → RHPP	GB	91/91	1.00 [0.93, 1.07]
BPA HZ1 → BPA HZ1	US	11/11	1.00 [0.89, 1.12]
BPA HZ2 → BPA HZ2	US	6/6	1.00 [0.81, 1.23]
<i>Between studies, same country and climate zone</i>			
Kloten → Hg	CH	50/21	0.97 [0.92, 1.01]
Kloten → MqO	CH	50/6	0.90 [0.83, 0.96]
BPA HZ2 → NREL	US	12/10	0.96 [0.91, 1.01]
NREL → BPA HZ2	US	10/12	1.05 [1.00, 1.12]
<i>Between countries, same climate class (Cfb)</i>			
Kloten → Hamelin	CH→DE	50/37	0.88 [0.85, 0.92]
<i>Between HP technologies</i>			
RHPP ASHP → RHPP GSHP	GB	137/45	1.14 [1.12, 1.20]
RHPP GSHP → RHPP ASHP	GB	45/137	0.87 [0.84, 0.90]
<i>Between social and private housing</i>			
RHPP social → RHPP private	GB	128/54	1.65 [1.62, 1.70]
RHPP private → RHPP social	GB	54/128	0.61 [0.58, 0.62]
<i>Between climate zones</i>			
BPA HZ1 → BPA HZ2	US	22/12	0.60 [0.57, 0.63]
BPA HZ2 → BPA HZ1	US	12/22	1.67 [1.60, 1.76]

to $\pm 7\%$ at 91. The same holds between studies of one country and climate zone: the sensitivity transfers from the BPA HZ2 households to the NREL households within 5%, and from Kloten to Hg and MqO within 10%. Between Kloten and Hamelin, two countries of the same climate class, it transfers within 12%.

The transfer degrades along three further axes. Between air- and ground-source HPs, the sensitivity differs by 13 to 14%. Between the two BPA heating zones, it differs by a factor of 1.7. Between social and private housing within the RHPP trial, it differs by a factor of 1.65, far more than between the two technologies. This is possibly because the two housing types differ in how their HPs are sized against the heat demand of the building, which sets how much of its capacity a population draws at a given temperature. A pilot therefore needs to match the target not only in climate and TCL technology, but also in housing stock.

VI. DISCUSSION

A. Accuracy

The capacity estimate carries two error sources. The first is the non-TCL temperature response that (1) neglects, which biases the net-load numerator upwards. A zero-HP control makes this concrete. Aggregates of 10 to 50 Kloten households without an HP still show a small heating arm, from which (10) infers a median 0.28 kW of spurious capacity per household. This floor is about 4% of the mean installed capacity of one HP, 6.3 kW. It therefore adds about 45% to the estimate at 10% HP penetration, but only about 4% once every consumer has an HP.

The second source is the transferred sensitivity. Within the Kloten population, the pilot's sensitivity matches the aggregates' own at a median ratio of 1.00, with an interquartile range of $\pm 10\%$, so the transfer adds spread rather than bias. Table IV sizes the two sources on the Kloten aggregates: with each aggregate's own SF sensitivity, the estimate errs at a WAPE of 14.5%, and with the pilot's at 19.3%.

On the real aggregates outside the pilot, where every household has an HP, the estimates lie within 10% of the submetered capacity.

B. Generalisation

The SF generalises in a specific and testable way. Within a population, its sensitivity is empirically stable against the number of HPs and their installed capacity over the tested range, so a curve from a submetered pilot serves that population's aggregates without submetering. Across populations, only the sensitivity transfers. The threshold temperature follows the local building stock and climate, and the net-load fit supplies it from data the DSO already meters. The intercept describes weather-independent TCL use, which lies outside the temperature-driven quantities estimated here. Transferring the full curve instead of the sensitivity alone raises the capacity error to 20% at MqO and 26% at Hamelin.

The sensitivity itself transfers between populations that share a climate zone and a housing stock. Between random splits of one population, between studies of one climate zone and between countries of one climate class, it stays within 12% of the target's own (Table VI). Between heating zones and between social and private housing, however, it differs by factors of 1.7 and 1.65, more than between HP technologies. Across the climate classes of Table II, it varies by up to a factor of four. Generalisation therefore reduces to transferring a single parameter from a pilot of the same climate zone, TCL technology and housing stock. This mirrors the inverse problem solved for PV installed capacity against irradiance [26], here posed against ambient temperature.

C. Data Requirements

The two fits ask for different data. The net-load fit needs only the aggregated net load and ambient temperature, both already held by the DSO, and runs on any aggregate. The SF fit needs submetered TCL load and installed capacity, which the DSO rarely holds. A single submetered pilot of the same population calibrates it, and its sensitivity then transfers, so no aggregate under study is submetered. A pilot of a few tens of HPs appears to suffice: a pilot of 25 Kloten HPs reproduces the SF shape of the other 25 at a median R_h^2 of 0.91, and recovers their capacity at a median error of +8% (Section IV-C). The pilot should, however, submeter every electric heating source of its households, including backup resistance heat.

D. Limitations

Several limits temper the results. The within-population tests rest on the 50 submetered Kloten HPs. The semi-synthetic aggregates are drawn without replacement and never share an HP with their pilot, but aggregates of the same split draw on the same 25 HPs and therefore overlap. The cross-population tests of the full estimator rest on three real aggregates in the heating regime, one of them with only six HPs, and all with an HP in every household. Table VI tests the SF transfer alone on further populations, but not the net-load numerator. The cooling regime remains untested. The only aggregates with complete cooling ground truth, the NEEA households of Washington and Oregon [52], allow no cooling transfer, since the Washington pilot has no day above its cooling threshold in 2023. The Oregon aggregate also shows that the numerator can fail on its own: even with its own SF sensitivities, its capacity is overestimated by 45% in heating and 72% in cooling. About a third of its net-load heating sensitivity lies outside the metered heating circuits. The cooling fits of Table II are therefore descriptive only. The model neglects the non-TCL temperature response throughout, which the bias at a low HP count exposes.

A further confounder is deliberate control. Swiss and German domestic HPs commonly run under interruptible tariffs, under which the utility may cut the HP supply during defined blocking periods, and since January 2024 §14a EnWG lets German DSOs throttle HPs above 4.2 kW [60]. Such control truncates the aggregated TCL peak. It may bias the net-load numerator and the capacity estimate downwards, and imprints regime-specific structure that a single-sensitivity transfer need not capture. On the real aggregates, however, the estimates lie within 10% of the submetered capacity, so such control is not a visible source of error there. The WPuQ campaign predates §14a, but blocking-period tariffs do not. Underlying all of these is the broader shortage of submetered TCL data, which a wider pilot would relieve.

VII. CONCLUSION

This paper presented a method to estimate the installed capacity of the TCLs behind a feeder from its net load and ambient temperature. Below a balance-point temperature, the daily net load and the SF both rise linearly as the temperature falls. The ratio of the two slopes gives the installed capacity, and only the SF slope is calibrated on a small set of submetered households. On 1260 aggregates built from measured Swiss households, the capacity was recovered at a WAPE of 19%, falling to 12–14% for aggregates of fifty households at 50% HP penetration. Most of this error stems from the net-load slope, which also includes the temperature response of households without an HP, rather than from the transfer of the SF slope. On three real Swiss and German aggregates where every household has an HP, the estimates came within 10% of the submetered capacity, whereas transferring the full calibrated SF curve erred by up to 26%. Across submetered households in five countries, the SF slope transferred within one climate zone but not between climate zones or between social and private housing. The calibration households should therefore match the target in climate, TCL technology and housing stock. The same fits give the temperature-driven energy without any submetered households, at a WAPE of 9%.

The method thus trades a modest, characterised error for a large gain in deployability. Its estimates are least reliable on aggregates with a handful of HPs and most reliable once tens of HPs are aggregated, which is the regime a substation presents. Where no submetered households are available, the net-load fit alone still gives the temperature-driven energy.

Regarding future work, four directions follow. The submetered populations were small, so a wider and more diverse set of metered HPs, with every electric heating source submetered, would test the transfer on more populations and in the cooling regime. The net-load sensitivity also absorbs the temperature response of the non-TCL load, which inflates the capacity estimate most where consumers without TCLs dominate. Future work will compensate the estimates for this response, for instance by subtracting a per-consumer non-TCL sensitivity scaled by the consumer count the DSO already holds. It will also evaluate data-driven methods for HP and TCL quantification against the proposed estimator, to see where a learned model earns its data cost and where the proposed estimator suffices. Finally, it will model the TCL consumption that does not depend on temperature, chiefly domestic hot water and baseline operation. The temperature-driven estimate omits this consumption, which sets the gap between the temperature-driven and the annual energy.

REFERENCES

- [1] International Energy Agency, “The Future of Heat Pumps,” IEA, Paris, Tech. Rep., 2022, [Online]. Available: <https://www.iea.org/reports/the-future-of-heat-pumps>.
- [2] M. Treutlein, P. Bothe, M. Schmidt, R. Hahn, O. Neumann, R. Mikut, and V. Hagenmeyer, “Real-world energy data of 200 feeders from low-voltage grids with metadata in Germany over two years,” *arXiv:2602.03521*, 2026.
- [3] E. Veldman, M. Gibescu, H. Slootweg, and W. L. Kling, “Impact of electrification of residential heating on loading of distribution networks,” in *Proc. IEEE Trondheim PowerTech*, 2011.
- [4] M. Akmal, B. Fox, J. D. Morrow, and T. Littler, “Impact of heat pump load on distribution networks,” *IET Gener. Transm. Distrib.*, vol. 8, no. 12, pp. 2065–2073, 2014.
- [5] C. van Someren, M. Visser, and H. Slootweg, “Impacts of electric heat pumps and rooftop solar panels on residential electricity distribution grids,” in *2021 IEEE PES Innovative Smart Grid Technologies Europe (ISGT-Europe)*. IEEE, 2021, pp. 1–6.
- [6] T. Becker, R. Smet, B. Macharis, and K. Vanthournout, “Effect of electric vehicles, heat pumps, and solar panels on low-voltage feeders: Evidence from smart meter profiles,” *Sustainable Energy, Grids and Networks*, vol. 42, p. 101705, 2025.
- [7] R. Liao, M. Manfren, P. A. B. James, and T. Zang, “Quantifying the impact of residential heat pumps on primary substations under critical cold events: A UK case study,” *Energy*, 2026.
- [8] A. M. V. Gouveia, M. U. Hashmi, R. D’hulst, and D. Van Hertem, “Importance of Aggregated DER Installed Capacity in Distribution Networks,” *arXiv preprint arXiv:2604.13926*, 2026.
- [9] H. Fei, Y. Kim, S. Sahu, M. Naphade, S. K. Mamidipalli, and J. Hutchinson, “Heat pump detection from coarse grained smart meter data with positive and unlabeled learning,” in *Proc. 19th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*. ACM, 2013, pp. 1330–1338.
- [10] A. Weigert, K. Hopf, N. Weinig, and T. Staake, “Detection of heat pumps from smart meter and open data,” *Energy Informatics*, vol. 3, no. S1, 2020.
- [11] T. Brudermueller, F. Breer, and T. Staake, “Disaggregation of Heat Pump Load Profiles From Low-Resolution Smart Meter Data,” in *Proc. 10th ACM Int. Conf. Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys)*. ACM, 2023, pp. 228–231.
- [12] O. Gisiger, A. Melillo, and P. Schuetz, “Heat pump detection and load disaggregation in low-resolution smart meter data with convolutional neural networks,” *Energy and AI*, p. 100691, 2026.
- [13] A. F. Moreno Jaramillo, D. M. Lavery, D. J. Morrow, J. Martinez del Rincon, and A. M. Foley, “Effective identification of distributed energy resources using smart meter net-demand data,” *IET Smart Grid*, vol. 5, no. 5, pp. 315–328, 2022.
- [14] S. Wang, R. Li, A. Evans, and F. Li, “Regional nonintrusive load monitoring for low voltage substations and distributed energy resources,” *Applied Energy*, vol. 260, p. 114225, 2020.
- [15] E. O’Malley, D. Pugh, J. Wu, and D. Clements, “A Deep Learning-Based Method for Non-Intrusive Load Monitoring and Load Disaggregation of 11kV/400V Electrical Substations,” in *ASME 2024 18th Int. Conf. Energy Sustainability*. ASME, 2024.
- [16] G. S. Ledva, L. Balzano, and J. L. Mathieu, “Real-time energy disaggregation of a distribution feeder’s demand using online learning,” *IEEE Trans. Power Syst.*, vol. 33, no. 5, pp. 4730–4740, 2018.
- [17] M. T. Ali, M. Mokhtar, M. Chiesa, and P. Armstrong, “A cooling change-point model of community-aggregate electrical load,” *Energy and Buildings*, vol. 43, no. 1, pp. 28–37, 2011.
- [18] K. X. Perez, K. Cetin, M. Baldea, and T. F. Edgar, “Development and analysis of residential change-point models from smart meter data,” *Energy and Buildings*, vol. 139, pp. 351–359, 2017.
- [19] N. MacMackin, L. Miller, and R. Cariveau, “Modeling and disaggregating hourly effects of weather on sectoral electricity demand,” *Energy*, vol. 188, p. 115956, 2019.
- [20] J. L. Mathieu, S. Koch, and D. S. Callaway, “State estimation and control of electric loads to manage real-time energy imbalance,” *IEEE Trans. Power Syst.*, vol. 28, no. 1, pp. 430–440, 2013.
- [21] H. Hao, B. M. Sanandaji, K. Poolla, and T. L. Vincent, “Aggregate flexibility of thermostatically controlled loads,” *IEEE Trans. Power Syst.*, vol. 30, no. 1, pp. 189–198, 2015.
- [22] X. Zhang and S. Grijalva, “A data-driven approach for detection and estimation of residential PV installations,” *IEEE Transactions on Smart Grid*, vol. 7, no. 5, pp. 2477–2485, 2016.
- [23] K. Li, F. Wang, Z. Mi, others, and T. Wang, “Capacity and output power estimation approach of individual behind-the-meter distributed photovoltaic system for demand response baseline estimation,” *Applied Energy*, vol. 253, p. 113595, 2019.
- [24] F. Sossan, L. Nespoli, V. Medici, and M. Paolone, “Unsupervised disaggregation of photovoltaic production from composite power flow measurements of heterogeneous prosumers,” *IEEE Transactions on Industrial Informatics*, vol. 14, no. 9, pp. 3904–3913, 2018.

- [25] L. Waswa, M. J. Chihota, and B. Bekker, "A probabilistic estimation of PV capacity in distribution networks from aggregated net-load data," *IEEE Access*, vol. 9, pp. 140 358–140 371, 2021.
- [26] A. M. Gouveia, M. U. Hashmi, R. D'hulst, and D. Van Hertem, "Installed pv capacity detection on lv substations: Comparison of data-driven and model-based methods," *International Journal of Electrical Power & Energy Systems*, vol. 177, p. 111848, 2026. [Online]. Available: <https://doi.org/10.1016/j.ijepes.2026.111848>
- [27] J. A. Azzolini, M. J. Reno, and J. Yusuf, "A model-free approach for estimating service transformer capacity using residential smart meter data," *IEEE J. Photovolt.*, vol. 14, no. 1, pp. 65–73, 2024.
- [28] J. Love, A. Z. P. Smith, S. Watson, E. Oikonomou, A. Summerfield, C. Gleeson, P. Biddulph, L. F. Chiu, J. Wingfield, C. Martin, A. Stone, and R. Lowe, "The addition of heat pump electricity load profiles to GB electricity demand: Evidence from a heat pump field trial," *Applied Energy*, vol. 204, pp. 332–342, 2017.
- [29] M. Chesser, P. O'Reilly, P. Lyons, and P. Carroll, "The impact of extreme weather on peak electricity demand from homes heated by air source heat pumps," *Energy Sources, Part B: Economics, Planning, and Policy*, vol. 16, no. 8, pp. 707–718, 2021.
- [30] A. Anderson, B. Stephen, R. Telford, and S. McArthur, "A Probabilistic Model for Characterising Heat Pump Electrical Demand versus Temperature," in *2020 IEEE PES Innovative Smart Grid Technologies Europe (ISGT-Europe)*. IEEE, 2020, pp. 1030–1034.
- [31] A. Anderson, B. Stephen, R. Telford, and S. D. J. McArthur, "Predictive thermal relation model for synthesizing low carbon heating load profiles on distribution networks," *IEEE Access*, vol. 8, pp. 195 290–195 304, 2020.
- [32] F. L. Müller and B. Jansen, "Large-scale demonstration of precise demand response provided by residential heat pumps," *Appl. Energy*, vol. 239, pp. 836–845, 2019.
- [33] R. D'hulst, W. Labeuw, B. Beusen, S. Claessens, G. Deconinck, and K. Vanthournout, "Demand response flexibility and flexibility potential of residential smart appliances: Experiences from large pilot test in Belgium," *Appl. Energy*, vol. 155, pp. 79–90, 2015.
- [34] G. Le Ray, M. H. Christensen, and P. Pinson, "Detection and Characterization of Domestic Heat Pumps," in *2019 IEEE Milan PowerTech*. IEEE, 2019, pp. 1–6.
- [35] M. F. Fels, "Energy conservation scorekeeping: The PRISM method," *AIP Conf. Proc.*, vol. 135, pp. 169–183, 1985.
- [36] D. Ruch and D. E. Claridge, "A Four-Parameter Change-Point Model for Predicting Energy Consumption in Commercial Buildings," *J. Solar Energy Eng.*, vol. 114, no. 2, pp. 77–83, 1992.
- [37] J. K. Kissock, J. S. Haberl, and D. E. Claridge, "Inverse modeling toolkit: Numerical algorithms," *ASHRAE Trans.*, vol. 109, no. 2, pp. 425–434, 2003.
- [38] R. Fazeli, M. Ruth, and B. Davidsdottir, "Temperature response functions for residential energy demand – a review of models," *Urban Climate*, vol. 15, pp. 45–59, 2016.
- [39] B. Liu, J. Tian, W. Luan, Y. Gao, X. Wang, and S. Luo, "Residential PV capacity estimation and power disaggregation using net metering measurements," *Global Energy Interconnection*, vol. 5, no. 6, pp. 590–603, 2022.
- [40] A. M. V. Gouveia, R. D'hulst, M. U. Hashmi, and D. Van Hertem, "Data-driven photovoltaic installed capacity detection at low voltage substation level," in *14th Mediterranean Conference on Power Generation Transmission, Distribution and Energy Conversion (MEDPOWER 2024)*, 2024, pp. 678–683.
- [41] G. Rouwhorst, A. Pondes, P. H. Nguyen, and H. Slootweg, "Medium-term forecasting of daily aggregated peak loads from heat pumps using clustering-based load duration curves to calculate the annual impact on medium to low voltage transformers," *IET Smart Grid*, vol. 7, no. 2, pp. 157–171, 2023.
- [42] C. Ziras, C. Heinrich, M. Pertl, and H. W. Bindner, "Experimental flexibility identification of aggregated residential thermal loads using behind-the-meter data," *Appl. Energy*, vol. 242, pp. 1407–1421, 2019.
- [43] D. S. Callaway, "Tapping the energy storage potential in electric loads to deliver load following and regulation, with application to wind energy," *Energy Convers. Manage.*, vol. 50, no. 5, pp. 1389–1400, 2009.
- [44] P. Westermann, C. Deb, A. Schlueter, and R. Evins, "Unsupervised learning of energy signatures to identify the heating system and building type using smart meter data," *Applied Energy*, vol. 264, p. 114715, 2020.
- [45] Pecan Street Inc., "Dataport: Residential Electricity and Water Data," [Online]. Available: <https://dataport.pecanstreet.org>, 2018, accessed: 2026-09-05.
- [46] H. Fani, M. U. Hashmi, and G. Deconinck, "Impact of electric vehicle charging simultaneity factor on the hosting capacity of lv feeder," *Sustainable Energy, Grids and Networks*, vol. 40, p. 101581, 2024.
- [47] M. Schlemminger, T. Ohrdes, E. Schneider, and M. Knoop, "Dataset on electrical single-family house and heat pump load profiles in Germany," *Scientific Data*, vol. 9, no. 1, 2022.
- [48] T. Brudermueller, E. Fleisch, M. González Vayá, and T. Staake, "HEAPO – An Open Dataset for Heat Pump Optimization with Smart Electricity Meter Data and On-Site Inspection Protocols," in *Proc. 16th ACM Int. Conf. Future and Sustainable Energy Systems (e-Energy)*. ACM, 2025, pp. 699–711.
- [49] K. Kaiser, M. Kreft, M. G. Vayá, and G. Hug, "Dataset on residential electricity load profiles in switzerland," arXiv:2607.19294, 2026.
- [50] Å. L. Sørensen, S. K. Lien, and H. T. Walnum, "COFACTOR-residential: Hourly electricity and heating data from residential buildings in Norway," *Data in Brief*, vol. 65, p. 112580, 2026, dataset 1 (Oslo/Bærum apartment blocks), CC BY 4.0. [Online]. Available: <https://doi.org/10.60609/3ab7-ez93>.
- [51] E. J. H. Wilson, A. Parker, A. Fontanini, E. Present, J. L. Reyna *et al.*, "End-Use Load Profiles for the U.S. Building Stock: Methodology and Results of Model Calibration, Validation, and Uncertainty Quantification," National Renewable Energy Laboratory (NREL), Tech. Rep. NREL/TP-5500-80889, 2022, resStock EUSS 2024.2 release. [Online]. Available: https://data.openei.org/s3_viewer?bucket=oedi-data-lake.
- [52] Northwest Energy Efficiency Alliance, "Northwest End-Use Load Research Project: Home Energy Metering Study (HEMS) 15-minute Data," [Online]. Available: <https://neea.org/data/nw-end-use-load-research-project>, 2025.
- [53] H. E. Beck, N. E. Zimmermann, T. R. McVicar, N. Vergopolan, A. Berg, and E. F. Wood, "Present and future köppen-geiger climate classification maps at 1-km resolution," *Scientific Data*, vol. 5, no. 1, p. 180214, 2018.
- [54] N. Chapman, L. Zhang, N. Good, and P. Mancarella, "Exploring flexibility of aggregated residential electric heat pumps," in *Proc. IEEE Int. Energy Conf. (EnergyCon)*. IEEE, 2016, pp. 1–6.
- [55] S. Few, P. Djapic, and G. Strbac, "A geographically disaggregated approach to integrate low-carbon technologies across local electricity networks," *Nature Energy*, vol. 9, no. 7, pp. 871–882, 2024.
- [56] Q. Meng, M. Mourshed, and S. Wei, "Going beyond the mean: Distributional degree-day base temperatures for building energy analytics using change point quantile regression," *IEEE Access*, vol. 6, pp. 39 532–39 540, 2018.
- [57] Department for Business, Energy and Industrial Strategy, "Renewable Heat Premium Payment Scheme: Heat Pump Monitoring: Cleaned Data, 2013–2015," UK Data Service, SN 8151, 2017, [Online]. Available: <https://datacatalogue.ukdataservice.ac.uk/studies/study/8151>.
- [58] B. Larson, J. Bush, R. Hong, and B. Davis, "High-Performance High-Capacity Heat Pump Field Study Report," Bonneville Power Administration, Portland, OR, USA, Tech. Rep., 2026, data: U.S. DOE Heat Pump Database. [Online]. Available: <https://heatpumpdata.energy.gov>.
- [59] J. Winkler and S. Ramaraj, "Field Validation of Air-Source Heat Pumps for Cold Climates," National Renewable Energy Laboratory (NREL), Golden, CO, USA, Tech. Rep. NREL/TP-5500-84745, 2023, data: U.S. DOE Heat Pump Database. [Online]. Available: <https://heatpumpdata.energy.gov>.
- [60] Bundesnetzagentur, "Festlegung zur netzorientierten Steuerung von steuerbaren Verbrauchseinrichtungen und steuerbaren Netzanschlüssen nach §14a EnWG (BK6-22-300)," Bundesnetzagentur, Bonn, Germany, Tech. Rep., 2023.